

PABLO NEIRZ VERGARA

SENIOR DATA SCIENCE ENGINEER

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PROFESSIONAL SUMMARY

Senior data scientist with a PhD in Computer Science from Universidad Técnica Federico Santa María (UTFSM) and over 10 years of experience in Python, machine learning, natural language processing (NLP), computer vision, and big data analytics. Research focused on trustworthy AI and causal machine learning, with publications in high-impact journals including Scientific Reports (Nature Portfolio) and Algorithms (MDPI). In industry, specialized in optimizing e-commerce operations and forecasting with state-of-the-art models such as TFT, NHITS, NBEATS, and LightGBM, consistently achieving forecast accuracies above 90%. Skilled at communicating complex findings to cross-functional teams: 1st and 3rd place finishes in international data science competitions, keynote speaker at events such as FLISOL, and featured on the ProinChile podcast on Spotify. Strong command of NLP techniques (embeddings and semantic similarity between concepts), prompt engineering with rules and MCP, pattern detection, and computer vision (image segmentation, OCR, object detection).

EDUCATION

PHD IN COMPUTER SCIENCE

Universidad Técnica Federico Santa María, 2022 - 2026

MASTER'S IN NEUROSCIENCE

Universidad de Oviedo / Autónoma de Chile, 2020 - 2022

MASTER'S IN MANAGEMENT AND EXECUTIVE DEVELOPMENT

Universidad de Nebrija, 2020 - 2021

MASTER'S IN BIG DATA

Cerem Global Business School, 2018 - 2020

BACHELOR OF SCIENCE IN PHYSICS

Universidad de Concepción, 2011 - 2017

PUBLICATIONS

Epistemic Frontiers and the Distinction between Causality, Information, and Predictability in Pattern Recognition

Pablo Neirz Vergara, Héctor Allende, Carolina Saavedra. *Scientific Reports (Nature Portfolio)*, 2026. [DOI: 10.1038/s41598-026-52883-z](#)

Attribute Relevance Score: A Novel Measure for Identifying Attribute Importance

Pablo Neirz Vergara, Héctor Allende, Carolina Saavedra. *Algorithms (MDPI)*, 2024. [DOI: 10.3390/a17110518](#)

Prediction of Concrete Compressive Strength through Artificial Neural Networks

Pablo Neirz Vergara, Leonardo Bennun, Mauricio Pradena, J. Gómez. *Građevinar*, 72(7), 585–592, 2020. [DOI: 10.14256/JCE.2438.2018](#)

Guide: Non-Linear Feature Generation with GPlearn

Pablo Neirz Vergara. *Kaggle (Bronze Medal)*, 2024. [Link](#)

Dyslexia Detection using Convolutional Neural Networks (Dual Master's Thesis)

Pablo Neirz Vergara. *Universidad de Oviedo / Universidad Autónoma de Chile*, 2022.

PROFESSIONAL EXPERIENCE

UNIVERSIDAD TÉCNICA FEDERICO SANTA MARÍA | PART-TIME PROFESSOR

Chile, Present

- Part-time teaching at Universidad Técnica Federico Santa María.

MERCADOLIBRE | SENIOR DATA SCIENCE ENGINEER

Santiago, Chile, December 2023 - Present

- Design and implement autonomous AI labs to improve machine learning algorithm performance, test hypotheses reproducibly, and document both effective approaches and those that do not yield improvements
- Develop advanced machine learning models to optimize e-commerce operations, improving inventory and sales forecasting accuracy with state-of-the-art models such as TFT, NHITS, NBEATS, and LightGBM
- Collaborate with international teams to define priority air cargo strategies and develop prioritization models, achieving forecast accuracies consistently above 90% and increasing logistics efficiency by 20%
- Developed and implemented a machine learning ensemble model for time series, adopted by multiple teams, that outperformed Meta's Prophet model by up to 10 percentage points in forecast accuracy, significantly improving data-driven decision making

TOTTUS | SENIOR DATA SCIENTIST

Santiago, Chile, April 2022 - December 2023

- Led the development of predictive models for national and imported product sales, improving accuracy in promotions, coupons, and store distribution, with errors below 5% MAPE
- Directed the Catman forecasting team, delivering analytical solutions that optimized product placement and sales strategies in new sales centers in Peru
- Presented actionable insights to stakeholders, connecting technical and business teams

SOOFT TECHNOLOGY | SENIOR DATA SCIENTIST (CONSULTANT FOR SAS)

Santiago, Chile, March 2022 - May 2022

- Monitored and validated credit risk models, preparing detailed performance reports
- Developed customer prospecting models, improving acquisition strategies

CAPITARIA | DATA SCIENTIST

Santiago, Chile, November 2020 - March 2022

- Built a qualitative model to analyze company operations, identifying growth opportunities and creating KPIs to monitor executive performance
- Led research on customer behavior, delivering hyper-personalized dashboards that reduced customer churn by 30% and increased profitability
- Wrote two articles for the company's official blog, explaining key findings in an accessible manner. [Link](#).

MINISTRY OF TRANSPORTATION AND TELECOMMUNICATIONS, CHILE | DATA SCIENTIST

Santiago, Chile, November 2020 - March 2022

- Led the technical counterpart in public tenders, including the 'Foreign Trade Logistics Barometer 2020' project. [Link](#).
- Developed port cargo prediction models with a mean squared error below 1%, delivered directly to the Transportation Undersecretariat

AVLA | DATA SCIENTIST

Santiago, Chile, March 2019 - July 2020

- Served as interim lead of the analytics area for Chile and Peru, reorganizing teams after the impact of COVID using agile methodologies
- Developed scrapers for public market data and used transparency law data to create a tender alert system, achieving a competitive advantage of several months
- Developed OCR systems to automate visual verification (captcha) steps in public data collection
- Developed NLP models to detect patterns in public markets

MEDICAL IMAGE ANALYSIS, UDEC | RESEARCHER

Concepción, Chile, March 2018 - December 2019

- Conducted brain connectivity analysis through the combined use of functional MRI and tractography (FONDECYT project ACT 1161427)
- Developed an ant colony algorithm for image edge detection, applied to artifact segmentation in brain MRI

APPLIED PHYSICS LABORATORY, UDEC | RESEARCH ASSISTANT

Concepción, Chile, March 2015 - March 2016

- Developed predictive models for concrete compressive strength using multilayer perceptrons optimized with grid search
- Analyzed heavy metals in drinking water using X-ray fluorescence techniques

COMPETITIONS & AWARDS

MercadoLibre Innovation Challenge 2024 — Winner

MercadoLibre

Financial Inclusion in Latin America — 1st Place

[datasource.ai](#) | [Link](#)

Apartment Price Prediction in Argentina and Colombia — 3rd Place

[datasource.ai](#) | [Link](#)

Flight Delay Prediction — 11th Place

[datasource.ai](#) | [Link](#)

AI Agents - Rock, Paper, Scissors — Top 25%

Kaggle

CERTIFICATIONS

- English C1 Advanced (EF SET Certificate) — EF Standard English Test (2026)
- SAS Essentials — SAS (2022)
- Advanced Growth Strategies — Reforge (2022)
- Behavioral Finance — Duke University (Coursera) (2021)
- Technical Analysis and Trading Strategy — Andia Academy (2021)
- Growth Hacking — SENCE (2020)
- Big Data Visualization — SENCE (2020)
- TensorFlow Workshop — Metric Learning (2019)
- Project Management with Agile Methodologies — SENCE (2019)
- SQL Fundamentals — SoloLearn (2018)

- Basic Concepts of Strategic Planning — Universidad Andrés Bello (2018)
- Medical Image Analysis — Universidad de Concepción (2018)
- Data Structures and Algorithm Analysis — Universidad de Concepción (2018)

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML

Python Libraries: NumPy, Pandas, Polars, Dask, Ibis, Scikit-learn, MLforecast, Sktime, GPlearn, TensorFlow, LightGBM, XGBoost, CatBoost

Visualization and Analysis: Plotly, TA-Lib, Seaborn, Matplotlib, Optuna

Cloud and Databases: GCP, BigQuery, SQL

Tools and Platforms: Figma, Jira, Cursor, Jupyter Labs

Specialization Domains: Machine Learning, Deep Learning, Big Data, Statistical Analysis, Neuroscience, Consumer Behavior, Forecasting, Agile Methodologies, NLP, Computer Vision, Medical Image Analysis, Causal ML, Trustworthy AI

LANGUAGES

- Spanish: Native
- English: Advanced (C1, EF SET certified)